
COMPANION WORKBOOK

PM Is Now Another Member of Technical Staff **Workbook**

Twenty chapter exercises mapped to twenty-five reusable product records.

Razii Abraham · First edition · 2026 · A4 print edition

Workbook Guide

These worksheets turn the book's operating chain into reusable artifacts:

signal -> evidence -> decision -> specification -> system behavior -> review -> release -> measurement -> learning

Use one real, bounded problem across the workbook where possible. Link every artifact to its source evidence and successor rather than copying context manually. Delete fields that do not apply, but do not silently skip a risk or ownership question: record why it is not relevant.

The templates are decision aids, not mandatory process. Adapt them to the team's domain, risk, tools, and governance. They do not replace engineering, security, data, design, legal, accessibility, or behavioral-science review.

How to Use This Workbook

Choose one of three modes:

1. **Quick path:** Chapters 1, 8, 13, 15, and 20. Map your participation boundary, expose one flow, define an AI quality bar, create portable context, and carry one slice through delivery.
2. **Chapter path:** complete the mapped worksheet after each chapter. Reuse the same product problem where the fit is natural.
3. **Capstone path:** begin with one Assumption and Evidence Map, then complete only the artifacts required to carry that claim through prototype, implementation, review, release, and learning.

The PDF is designed for printing and handwriting. The Markdown files remain the editable source for teams that prefer to complete the records digitally.

Chapter-to-Workbook Map

Chapter	Exercise	Workbook record
1	Map your technical floor	Technical Floor Map
2	Turn a user story into an API conversation	API State Worksheet
3	Turn a feature idea into a release path	Feature Release Path Record
4	Rewrite and hand off an image requirement	Image Requirement Record
5	Turn a product question into SQL thinking	Product Data Question Record
6	Write a tracking plan	Tracking Plan
7	Design an experiment before results exist	Experiment Pre-Registration Card
8	Expose ambiguity in a flow	Flow-to-Acceptance Record
9	Audit one screen and one reusable component	Design Audit and Component Contract
10	Diagnose one target behavior twice	Behavior Intervention Card
11	Map the problem shape	Evidence Card and API State Worksheet
12	Compare solution shapes	AI Solution-Fit Decision Record
13	Design an AI evaluation system	AI Evaluation Scorecard
14	Design a calibration loop	Prompt Evaluation Table, AI Calibration

		Record, and Ship Decision Record
15	Create a portable context brief	Portable Context Brief
16	Build a repo orientation map	Repository Orientation Map
17	Create an assumption and evidence map	Assumption and Evidence Map
18	Write a prototype contract	Prototype Contract
19	Design a product instrument	Product Instrument Design Record and Productionization Memo
20	Take one slice through delivery	End-to-End Delivery Slice Record, PR Review Checklist, and Ship Decision Record

Artifact Index

The assembled workbook contains 25 reusable records. Some chapters use more than one record because the exercise crosses product, technical, or release boundaries. Reusing a record is intentional: the operating chain should carry evidence forward rather than duplicate it.

RECORD 1 OF 25

Technical Floor Map

Product Area and Promise

- Product area:
- User or operational actor:
- Product promise: what should they be able to trust?
- Decision this map should improve:

System Behavior Behind the Promise

State, system, data, or integration	Why the promise depends on it	Current evidence/source	Important unknown

Participation Level

Level	What I can do today	Evidence or example	Gap	Next safe practice
Ask				
Inspect				
Prototype				
Contribute				

Ownership Boundary

Decision or activity	My role	Accountable owner or specialist	When to involve them	Handoff evidence

Thirty-Day Practice Plan

Week	One action that follows the work further	Person or artifact needed	Evidence of progress
Week 1			
Week 2			
Week 3			
Week 4			

Reflection

- Where should I stop and ask for deeper expertise?

- What became clearer because I moved closer to the system?

- What will I inspect again after the next release?

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API State Worksheet

Product Promise

- User story: _____
- User action that starts the flow: _____
- Object/resource whose state changes: _____
- Decision this worksheet must clarify: _____

Actors and Contract

Actor/system	Request or event	Authentication/permission	Response or state change

Request

- Method/operation: _____
- Endpoint or contract reference: _____
- Required fields: _____
- Optional fields: _____
- Idempotency/retry identifier: _____
- Sensitive fields and handling boundary: _____

State Map

System state/status	User meaning	UI state/copy	Internal action	Support visibility	Event/metric

Failure and Recovery

Failure	How detected	Safe retry?	User recovery	Owner/escalation

Asynchronous Behavior

- Webhook/callback events: _____
- Source of truth: _____

- Late event behavior:

- Duplicate event behavior:

- Missing/out-of-order event reconciliation:

Acceptance and Verification

- Contract tests:

- Happy-path product check:

- Error/recovery checks:

- Security/logging checks:

- Analytics checks:

- Open engineering/security questions:

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Feature Release Path Record

Objective and Slice

- User problem: _____
- Intended outcome: _____
- Smallest buildable version: _____
- Explicitly excluded: _____
- Evidence supporting this slice: _____

Delivery Path

Stage	Product question	Artifact, environment, or check	Owner	Exit evidence
Brief and acceptance				
Local implementation				
Automated checks				
Review				
Preview or staging				
UAT				
Deployment				
User exposure				
Measurement				

Release Control

- Feature flag or cohort: _____
- Rollout stages: _____
- Pause condition: _____
- Rollback or disable method: _____
- Support and on-call visibility: _____
- Data or migration reversibility: _____

Product Verification

Scenario	Expected behavior	Where tested	Result/evidence	Owner/status
Happy path				
Validation failure				
Dependency failure				
Permission or role boundary				
Retry, duplicate, or recovery				

Learning Contract

- Primary outcome:

- Guardrail:

- Event, query, or dashboard:

- Review date:

- Continue, revise, pause, or stop rule:

- Questions still requiring engineering, design, data, security, or operational judgment:

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Image Requirement Record

User Job and Asset Role

- Product surface:
- User job:
- Why the image matters to trust, comprehension, or conversion:
- Asset owner and rights source:

Input Contract

Decision	Requirement	User-facing reason	Validation or fallback
Accepted formats			
Minimum dimensions			
Maximum file size			
Aspect ratio and crop			
Metadata retained			
Privacy or rights boundary			

Upload, Processing, and Storage

- Upload path and authorization:
- Original storage policy:
- Processing states:
- Generated variants:
- Retry and duplicate behavior:
- Retention and deletion:
- Failure ownership:

Delivery Contract

Context	Variant or transformation	Delivery path	Performance target	Fallback

User-Visible States

State	Trigger	Interface behavior and copy	Recovery	Event/evidence
Idle				
Uploading				
Processing				
Ready				
Failed				
Missing or unavailable				

Acceptance and Tradeoffs

- Required quality:
- Performance and cost guardrails:
- Accessibility requirement:
- Analytics or operational evidence:
- Open infrastructure or security decisions:
- What the requirement deliberately leaves to engineering/design judgment:

AI Coding-Agent Handoff

- Observed user problem:
- Smallest prototype slice needed to learn:
- Explicit non-goals:
- Assumptions the agent must surface:
- Tests, diff, screenshots, and other evidence to return:
- Human review gates before production architecture or deployment:

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Product Data Question Record

Question, Hypothesis, and Decision

- Product question:
- Hypothesis:
- Decision that could change:
- Evidence that would challenge the hypothesis:

Metric and Grain

- Metric or comparison:
- Object being counted, summed, or compared:
- One row in the intended result represents:
- Time window and time zone:
- Cohort or eligibility:

Source and Relationships

Table, collection, event, or dashboard	Relevant fields	Relationship/key	Source-of-truth concern

Query Plan

- Filters:
- Joins, lookups, or embedded paths:
- Grouping:
- Aggregation:
- Missing, duplicate, or late data behavior:

Write SQL, MongoDB syntax, or a pseudo-query here.

Validation and Limits

Check	Expected result	Actual evidence	Disposition
Row-count or grain check			
Duplicate check			
Null/missing-data check			
Dashboard or source comparison			
Sample-record inspection			

- What this analysis supports:
- What it cannot establish:
- Data or statistical specialist review needed:
- Next question:

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Tracking Plan

Product Decision

- Feature/journey:
- Assumption or decision:
- Primary outcome:
- Guardrails:
- Owner:

Event Contract

Event	Exact firing condition	Actor	Object	Required properties	Identity/time rule	Decision supported

Property Dictionary

Property	Type	Allowed values/source	Required?	Sensitive?	Meaning/version

Metric Definitions

Metric	Numerator	Denominator	Unit/grain	Window/time zone	Exclusions	Owner

Governance and QA

- Schema/version:
- Expected volume:
- Duplicate policy:
- Late event policy:
- Consent/retention/access constraints:
- Development validation:
- Production validation and alert:
- Backfill/reprocessing rule:

- **Retirement rule:**
-

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Experiment Pre-Registration Card

Decision

- Decision this experiment informs:

- Why a randomized experiment is appropriate:

- Decision deadline and reversibility:

Hypothesis and Mechanism

- Target population:

- Product change:

- Expected behavior:

- Why the change should cause it:

Design

- Randomization unit:

- Eligibility:

- Trigger condition and counterfactual logic:

- First possible treatment exposure:

- Estimand and analysis population:

- Exclusions:

- Bucketing moment:

- Stable assignment method:

- Traffic allocation:

- Control:

- Variant(s):

- Contamination/interference risk:

Measurement

- Primary metric and exact definition:

- Numerator, denominator, and metric grain:

- Baseline and source:

- Minimum effect worth acting on, absolute and relative:

- Treatment-effect guardrails and harm margins:

- Supporting metrics:

- Required events/properties:

Analysis Plan

- Statistical method and assumptions:

- Alpha/interval level and sidedness:

- Power and required sample for primary metric:

- Power or interval requirement for guardrails:

- Stopping rule and expected duration:

- Prespecified segments:

- Multiple-comparison treatment:

- Missing data/outlier policy:

Validity Checks

- Exposure logging:

- Sample-ratio mismatch:

- Pre-exposure balance diagnostics and interpretation:

- Instrumentation QA:

- Reasons to invalidate:

- Legitimate reasons for replication or follow-up:

Evidence Rule

- Supported benefit if:

- Supported harm if:

- Inconclusive if:

- Invalid if:

Product Decision Rule

- Ship if:

- Iterate if:

- Stop if:

- Roll back/avoid rollout if:

- Language required when evidence and product action differ:
-

Sign-Off

- Product:
 - Data/statistics:
 - Engineering:
 - Design/research:
 - Date locked before results:
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Flow-to-Acceptance Record

Scope and Operational Claim

- Product journey/object:
- User or operational problem:
- Start condition:
- End condition:
- In-scope implementation boundary:
- Explicitly deferred boundary and manual fallback:
- Source evidence and domain reviewers:

Actors and Ownership

Actor/system	Role in the flow	Actions allowed	Evidence produced	Handoff owner

Current State and Intended State

Step	Current behavior	Failure/cost	Intended behavior	Product/system affected

Object States and Transitions

From state	Trigger/action	Evidence required	To state	Custodian/owner	Invalid-transition behavior

Event and Audit Contract

- Stable object identifier:
- Actor and authorization evidence:
- Event and server timestamps:
- Location/device/application context:
- Transition-specific fields:
- Idempotency or duplicate rule:
- History and support visibility:

- SLA clock: starts, stops, pauses, and source timestamp:
-

Negative and Recovery Paths

Scenario	Detection	User/operator feedback	System behavior	Manual fallback/escalation	Owner
Duplicate action					
Offline or timeout					
Missing/invalid evidence					
Out-of-order event					
Disputed handoff					

Acceptance and Verification

ID	Given/when/then product behavior	UI/state check	API/data check	Event/analytics check	Owner/status

Open Decisions

Decision/question	Risk if unresolved	Decision owner	Due point	Resolution/source

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Design Audit and Remediation Record

Scope

- Product surface/journey:
- Users and tasks:
- Platforms/viewports:
- Audit methods:
- Design-system/version reference:
- Reviewers/date:

Findings

ID	Screen/component/state	Finding and evidence	User impact	Severity	Existing pattern/token	Remediation	Owner	Status

State Coverage

Component/flo w	Default	Loading	Empty	Error/recovery	Disabled	Success	Responsive	Keyboard/assis tive tech

Reuse Decision

- Existing component/pattern:
- Reuse, extend, or create:
- Product reason:
- Accessibility and interaction review:
- Code/design parity check:
- Maintainer/ownership decision:

Prioritization

- Critical before release:
- Next product slice:
- Design-system backlog:
- Accepted debt and expiry/review date:

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Design-System Component Contract

Purpose and Decision

- Component/pattern:
- User intent supported:
- Product surfaces and frequency:
- Existing system options reviewed:
- Decision: reuse, extend, or create:
- Why composition is or is not sufficient:
- Design owner / engineering owner:

Anatomy and Content

Element	Required/optional	Content rule	Interaction/semantic role	Truncation/absence behavior

API, Variants, and States

Prop/variant/state	Allowed values	Default	User-visible behavior	Invalid combination prevented?

Cover default, hover where relevant, pressed, focus, disabled, loading, empty, error, and success states. Delete states that genuinely do not apply and record why.

Token Contract

Decision	Semantic token	Component token, if justified	Raw value prohibited?	Figma variable/style	Code reference

Responsive and Compositional Behavior

- Supported widths/platforms:
- Minimum touch target:
- Reflow/wrapping rule:
- Long content/localization rule:

- Performance or rendering constraint:
-

Accessibility Contract

- Native semantic element/role:
 - Accessible name and description:
 - Keyboard order and activation:
 - Visible focus behavior:
 - Error/status announcement:
 - Contrast and non-color cues:
 - Reduced-motion/zoom/text-resize behavior:
 - Assistive-technology verification:
-

Figma-to-Code Parity

Contract area	Figma source/status	Code source/status	Verification	Drift owner/date
Variants and states				
Tokens				
Responsive behavior				
Accessibility				
Product usage				

Tests, Release, and Maintenance

- Unit/interaction tests:
 - Visual regression stories:
 - Keyboard/accessibility checks:
 - Product acceptance criteria:
 - Documentation and examples:
 - Version/release path:
 - Adoption or migration plan:
 - Deprecated alternatives and removal date:
 - Exceptions, owner, and review date:
-

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Behavior Intervention Card

Target Behavior and Context

- Specific observable behavior:
- User/audience and journey step:
- Decision risk: low, medium, or high:
- Current baseline or qualitative frequency:
- Why this behavior should create user value:

Evidence Ladder

Level	Claim	Evidence/source	Confidence and limitation
Observation			
Behavioral diagnosis			
Intervention hypothesis			
Outcome evidence			

Competing Diagnosis

Lens	Plausible barrier	Supporting evidence	Contradicting evidence	What would discriminate it?
Fogg: motivation				
Fogg: ability				
Fogg: prompt				
COM-B: capability				
COM-B: opportunity				
COM-B: motivation				

Intervention Contract

- Change proposed:
- Mechanism expected to change:
- User control: skip, revise, undo, or opt out:

- Safe default/fallback:
- Explanation and transparency:
- Accessibility and inclusion:
- Privacy/data collected, purpose, retention, and permission:
- Risk of manipulation, exclusion, or narrowing choice:

Evidence Plan

Evidence type	Metric/method	Segment/cohort	Decision threshold or interpretation rule	Owner
Mechanism				
Activation/outcome				
Qualitative				
System performance				
Longer-term directional signal				

Low-Traffic Plan

- Stable usability task and audience:
- Interviews/session review:
- Release-cohort comparison and caveats:
- Minimum repeated evidence before expanding:
- Claims this design cannot support:

Guardrails and Reversal

Potential harm/failure	Guardrail signal	Stop/revise condition	Response owner

- Evidence that would reverse the diagnosis:
- Rollback or safe-disable path:
- Decision date and result:
- What the team learned next:

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Evidence Card

- Evidence ID: _____
- Date observed: _____
- Owner: _____
- Decision or opportunity: _____

Source

- Source type: _____
- Source link or durable reference: _____
- Population/context: _____
- Collection method and period: _____
- Consent/confidentiality constraints: _____

Observation

State what was seen, heard, or measured without interpretation.

Interpretation

State the current explanation and plausible alternatives.

Strength and Limits

- Confidence: _____
- What this evidence supports: _____
- What it does not support: _____
- Known bias, missing context, or data-quality issue: _____

Decision Link

- Assumption affected: _____
- Decision changed or informed: _____
- Artifact/ticket link: _____

Next Test

- Smallest useful next question: _____

- Evidence that would increase confidence:
 - Evidence that would reverse the current interpretation:
-
-

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AI Solution-Fit Decision Record

Problem Without Technology

- Intended user and decision moment:
- Current behavior and missing capability:
- Approved data/content/tool asset:
- Consequence if the output is wrong:
- Required user control and fallback:
- Evidence/source links:

Problem Shape

Question	Evidence	Constraint/unknown
Which behavior must be exact?		
Which output may vary safely?		
Which interpretation creates value?		
What labels or ground truth exist?		
Which source must ground output?		
What should the system refuse or abstain from?		

Option Comparison

Dimension	No build/manual	Assisted workflow	Deterministic software	Predictive ML	Generative AI	Hybrid	Buy/integrate
User value							
Required data/labels							
Grounding/provenance							
Acceptable variability							
Failure consequence							
User control/fallback							
Cost/latency							

Smallest meaningful test							
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Responsibility Boundary

Responsibility	Deterministic code	Retrieval/provider	Predictive/generative component	UI/user	Human/domain owner
Eligibility and permissions					
Context construction					
Interpretation/generation					
Validation/abstention					
Storage/provenance					
Monitoring/escalation					

Decision

- Selected system shape and why:
- Rejected alternatives and evidence:
- Assumptions to test:
- Evaluation set required before release:
- Cost/latency boundary:
- Failure, fallback, and rollback:
- Privacy/security/rights review:
- Approvers/date:

AI Evaluation Scorecard

Capability and Release

- Capability/version:
- Model/prompt/tool/retrieval configuration:
- Intended users and task:
- Decision: prototype, internal test, pilot, expand, pause, or retire:

Evaluation Set

- Dataset/version:
- Source and consent:
- Representative slices:
- Edge/adversarial cases:
- Leakage check:
- Known coverage gaps:

Criteria

Criterion	Definition	Evaluator	Scale/threshold	Critical failure?

Results

Slice	Sample	Criterion results	Critical failures	Comparison to baseline

Evaluator Quality

- Human reviewer calibration:
- Model-judge agreement with humans:
- Disagreement review:
- Deterministic-check coverage:

Failure Taxonomy

Failure type	Count/rate	Severity	Likely layer	Owner	Regression case added?

Product and Risk Signals

- Task success/behavior metric:

- User feedback:

- Safety/privacy/fairness concerns:

- Latency and cost:

- Escalation/fallback behavior:

Decision Record

- Thresholds met/not met:

- Accepted residual risks:

- Required changes:

- Rollout boundary:

- Monitoring and recalibration trigger:

- Approvers/date:

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Prompt Evaluation Table

Capability

- Task and intended user value:
- Dataset/version:
- Model/tool/retrieval configuration:
- Baseline prompt/version:
- Candidate prompt/version:
- Evaluators and calibration date:

Case-Level Comparison

Case ID	Input/slice	Expected qualities	Baseline output/result	Candidate output/result	Criterion scores	Preferred	Failure tags	Reviewer note

Aggregate Results

Criterion/slice	Baseline	Candidate	Threshold	Delta	Confidence/limits

Regression and Risk Review

- Critical failures introduced:
- Previously failing cases fixed:
- Disagreements requiring adjudication:
- Cost/latency change:
- Safety/privacy change:
- Dataset leakage or overfitting concern:

Decision

- Adopt, revise, reject, or run further test:
- Evidence supporting decision:
- Residual risk:
- New regression cases:
- Version/release boundary:

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AI Calibration Record

Observed Failure

- Capability/version: _____
- User-visible failure and consequence: _____
- System/business consequence: _____
- Detection source: user report, event, trace, human review, judge, or regression: _____
- Severity and affected cohort: _____
- Preserved case/evidence location: _____

Version Chain

Layer	Current version/configuration	Evidence captured?
Input/context construction		
Retrieval/provider/index		
Entity resolution/ranking		
Prompt/model/sampling		
Tool and output schema		
Evaluator/rubric/judge		
Application/exposure		

Diagnosis

Candidate layer	Supporting evidence	Contradicting evidence	How to discriminate
Input/context			
Retrieval/provider			
Identity/ranking			
Prompt/orchestration			
Model capability			
Evaluation			

Policy/strategy			
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Bounded Change

- Diagnosed component:
- Smallest credible change:
- Expected mechanism:
- Layers deliberately unchanged:
- Cost/latency/privacy impact:
- New failure risk:

Comparison Plan

Case group	Cases/version	Required behavior	Critical regression	Evaluator
Preserved failures				
Normal regression				
Boundary/high risk				
Held out/novel				
Repeated-run stability				

Release and Production Learning

- Evaluation result and limits:
- Adopt, revise, reject, or investigate:
- Exposure boundary/feature flag:
- Fallback/rollback:
- Production sampling mix:
- Events/traces/qualitative evidence:
- Judge-to-human calibration check:
- Review owner/date:
- Evidence that would reverse the diagnosis:
- New regression cases and next decision:

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Portable Context Brief

Objective

- User or business problem:

- Desired decision or outcome:

- Why this matters now:

Evidence

Source	Observation	What it supports	Confidence/limits

Scope

- Included:

- Excluded:

- Smallest useful slice:

Product Behavior

- Entry condition:

- Happy path:

- Loading/pending states:

- Error and recovery states:

- Permissions and roles:

- Data created, read, changed, or deleted:

Relevant Surfaces

- Ticket/spec:

- Flow/design:

- Repository files or modules:

- Existing implementation precedent:

- API/schema/event documentation:

Constraints and Ownership

- Security/privacy/compliance:

- Design-system/accessibility:

- Performance/cost:

- Decisions the agent or contributor may make:

- Decisions requiring human owner approval:

Expected Output

- Artifact or change requested:

- Required explanation:

- Files or systems that must not change:

Verification

- Product checks:

- Automated checks:

- Data/event checks:

- Reviewers:

- Rollout/rollback expectations:

Open Questions

- _____

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Repository Orientation Map

Product Promise and Entry Point

- Product flow:
- User promise:
- User or system entry point:
- Read-only orientation prompt or task:

Rules and Reading Order

Priority	File, document, or instruction	What it governs	Verified date
1			
2			
3			

Product-to-Repository Map

Product behavior or state	Route/component/service/data	File reference	Tests/checks	Event/flag	Owner/reviewer

Delivery Surfaces

- Local run path:
- Preview or staging path:
- CI and required checks:
- Feature-flag location:
- Analytics convention:
- Related tickets, specs, decisions, and pull requests:

Evidence Classification

Claim	Verified fact, inference, contradiction, or unknown	Evidence	Consequence or next check

Orientation Boundary

What the repository proves

- ---

What the repository cannot answer

- ---

Human context required

- ---

Next Safe Action

- Smallest useful inspection, prototype, or change:

- Files or systems that must not change:

- Required reviewers:

- Verification before implementation:

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Assumption and Evidence Map

Product Bet and Original Signal

- Product bet or decision:
- Original observation:
- Observation type:
- Source and date:
- Current confidence and limits:

Artifact Lineage

Sequence	Artifact or conversation	Claim inherited	Claim changed or added	Evidence link
1				
2				
3				

Assumption Inventory

Assumption	Type	Uncertainty (1-5)	Consequence (1-5)	Priority	Current evidence
1					
2					
3					
4					
5					

Evidence Plan for the Top Assumptions

Assumption	Question	Evidence method	Population/source	Decision threshold	Owner/date

Evidence Update

Evidence item	Evidence for	Evidence against	Limit or alternative explanation	Confidence change

Decision

- Proceed, revise, narrow, pivot, stop, or investigate:

- Reason:

- Rejected alternatives:

- Next artifact or test:

- Evidence that would reverse this decision:

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Prototype Contract

Learning Question

- Decision the prototype informs:
- Riskiest assumption:
- What must be learned now:

Audience and Scenario

- Participant/user:
- Context and task:
- Starting information:
- Success behavior:

Fidelity

- Interaction fidelity:
- Content/data realism:
- AI-output realism:
- Visual/design-system fidelity:
- Technical integration fidelity:
- What is intentionally simulated:

Evidence Plan

- Method:
- Observable behaviors:
- Questions/prompts:
- Instrumentation:
- Sample and recruiting limits:
- Evidence that supports the assumption:
- Evidence that weakens or rejects it:

Boundaries

- Not safe or appropriate for:

- Claims stakeholders must not infer:

- Parts that cannot be reused as production code/content:

Stop Rule and Next Decision

- Stop when:

- Proceed if:

- Revise if:

- Abandon if:

- Owner and review date:

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Product Instrument Design Record

Decision and Input Boundary

- Recurring product decision:
- Intended users:
- Manual workflow today:
- Input source and owner:
- Permission, consent, rights, and retention boundary:
- No-output or abstention condition:

Structured Output

Field	Type/allowed values	Meaning	Evidence/provenance	Missing behavior

Pipeline

Stage	Input	Output	Quality question	Failure route	Owner
Scope and collection					
Ingestion and identity					
Extraction					
Taxonomy or mapping					
Evaluation					
Aggregation					
Presentation					

Evaluation Slice

Case	Input shape	Expected output or no-output	Risk/severity	Reviewer/result
1				
2				
3				

4				
5 - no-output case				

Error Taxonomy and Review

Error type	Detection	User/decision consequence	Human-review route	Correction path

Decision Surface

- Questions the dashboard, report, or export must answer:
- Required slices and filters:
- Observation versus inference treatment:
- Source and uncertainty display:
- User correction or challenge path:

Product Gate

Decision	Criteria	Evidence	Owner/date
Stop			
Improve prototype			
Productionize			
Pivot			

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Product Instrument Productionization Memo

Decision

- Instrument/capability:
- Decision requested:
- Users and decisions supported:
- Current proof-of-concept boundary:
- Recommended next state:

Input and Rights

- Data sources and owners:
- Consent/license/terms basis:
- Sensitive-data classification:
- Retention and deletion:
- Deduplication/identity/provenance:
- Prohibited uses:

Pipeline

Stage	Input/output contract	Implementation	Quality check	Failure route	Owner
Ingestion					
Extraction					
Taxonomy					
Mapping					
Aggregation					
Presentation/export					

Evaluation and Monitoring

- Evaluation dataset/version:
- Human review and calibration:
- Accuracy/quality dimensions:

- Slice and edge-case coverage:
- Release thresholds:
- Drift/data-quality alerts:
- Feedback and correction path:

Architecture and Operations

- Build versus buy decision:
- Model/prompt/tool versions:
- Latency/throughput target:
- Cost model and budget guardrail:
- Reliability/SLO/fallback:
- Observability and audit log:
- Security/threat review:
- Support/runbook/on-call ownership:

Rollout

- Internal/pilot/general-availability stages:
- Feature flag and cohort:
- User disclosure/explanation:
- Rollback/kill switch:
- Measurement and review date:

Open Risks and Sign-Off

Risk	Likelihood/impact	Mitigation	Owner	Due/status

- Product:
- Engineering:
- Data/AI:
- Security/privacy/legal:
- Operations/support:

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Product and PR Review Checklist

Context

- User problem and evidence:

- Ticket/spec/flow:

- PR/build/preview:

- Scope explicitly excluded:

Diff Orientation

- Files and systems changed:

- Product states changed:

- Data/schema/API/event changes:

- Existing pattern or reference implementation followed:

- Unexpected or unrelated changes:

Product Correctness

- Happy path matches acceptance criteria.

- Loading, empty, pending, error, permission, and recovery states are covered.

- Copy and interaction match the intended user context.

- Analytics events fire at the defined moment with valid properties.

- Feature flag, cohort, default, and fallback behavior are correct.

- AI output, if present, meets evaluation thresholds and routes failures safely.

Technical and Operational Quality

- Required format/type/lint/build/test checks pass.

- Security, privacy, logging, and secrets boundaries are respected.

- Design-system and accessibility patterns are reused correctly.

- Retry, duplication, idempotency, and concurrency risks are handled where relevant.

- Migration, backward compatibility, performance, cost, and monitoring are addressed.

- Rollback/disable path exists and is understood.

Verification Evidence

Check	Command/scenario	Result	Reviewer

Review Disposition

- Robot/read-only review findings:

- Product review findings:

- Engineer/design/data/security findings:

- Must fix before merge:

- Follow-up backlog:

- Approved by/date:

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Ship Decision Record

Change

- User problem and intended outcome:

- Scope included/excluded:

- Ticket/spec:

- Branch/PR/build:

- Owners:

Evidence and Acceptance

- Evidence supporting the decision:

- Required product behavior:

- Known limitations:

- Rejected alternatives:

Verification Record

Check	Result/evidence	Owner	Status
Happy path			
Error/recovery paths			
Permissions/data behavior			
Accessibility/design system			
Type/lint/build/tests			
Event and metric behavior			
AI evaluation, if applicable			
Robot/read-only review			
Human review			
Preview/staging			

Risk and Controls

- Feature flag/target cohort:

- Rollout stages:

- Guardrails and failure signals:

- Rollback/disable method:

- On-call/support visibility:

Release and GTM

- Release owner/date:

- Internal communication:

- Customer communication:

- Documentation/training:

Measurement and Learning

- Primary outcome:

- Guardrails:

- Dashboard/query:

- Review window:

- Continue/revise/pause rule:

- Result and next decision:

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End-to-End Delivery Slice Record

Pressure-Tested Brief

- User signal and source:
- Problem and intended outcome:
- Smallest useful slice:
- Expected behavior and acceptance:
- Exclusions and unresolved questions:
- Success and guardrail signals:

Planning Path

- Manual work, direct edit, prototype, or agent-assisted implementation:
- Why this path fits the risk and uncertainty:
- Repository precedents and rules:
- Files/systems likely to change:
- Files/systems explicitly protected:
- Required human owners and reviewers:

Implementation and Verification

Evidence	Result or link	Issue found	Disposition/owner
Local product QA			
Type/lint/build/tests			
Robot or read-only review			
Human technical review			
Product/design/data review			
Preview or staging			
Retest after review changes			

Exposure and GTM

- Deployment date/build:

- Feature flag and eligible cohort:
- Rollout stages:
- Customer/internal communication:
- Support and documentation:
- Pause, rollback, or kill-switch owner:

Boundary Memo

Boundary	Included now	Deferred/manual fallback	Risk and owner
Product behavior			
Data and instrumentation			
Reliability/operations			
Security/privacy/compliance			
AI/model behavior, if applicable			

Learning Record

- How will we know it worked?
- How will we know it caused harm?
- How will we know the team can maintain it?
- Measurement window and decision date:
- Continue, revise, pause, roll back, or stop rule:
- Result and next operating-loop decision: